Order of operations in SWIRL

Affine Transformations are *composable*. **SWIRL** performs them in the following order:

- P pivoting
- K skewing
- S scaling
- R rotation
- T translation

The mnemonic "*PiKe SoRTie*" simplifies memorizing it. Here we show the forms of these elemental transforms. When multiplied by a bi-dimensional *row vector* in the form $(x\ y\ 1)$, the product will yield the indicated result. Furthermore, the matrices can be multiplied together, so the vector only needs to be multiplied against the final matrix product.

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(% i1)  P: matrix([1, 0, 0], [0, 1, 0], [-p_x, -p_y, 1]);
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$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -p_x & -p_y & 1 \end{pmatrix} \qquad \text{(\% o1)}$$

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(% i2)  K: matrix([1, k_y, 0], [k_x, 1, 0], [0, 0, 1]);
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$$\begin{pmatrix} 1 & k_y & 0 \\ k_x & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \text{(\% o2)}$$

(% i3) S: matrix([s_x, 0, 0], [0, s_y, 0], [0,0,1]);

$$\begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad (\%\ o3)$$

(% i4) R: matrix([cos(θ), sin(θ), 0], [-sin(θ), cos(θ), 0], [0, 0, 1]);

$$\begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad (\%\ o4)$$

(% i5) T: matrix([1, 0, 0], [0, 1, 0], [t_x, t_y, 1]);

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{pmatrix} \qquad (\%\ o5)$$

Now for the cumulative transformations. Note that the "." operator means matrix multiplication in Maxima:

(% i6) P.K;

$$\begin{pmatrix} 1 & k_y & 0 \\ k_x & 1 & 0 \\ -k_x p_y - p_x & -p_y - k_y p_x & 1 \end{pmatrix} \qquad (\%\ o6)$$

(% i7) P.K.S;

$$\begin{pmatrix} s_x & k_y s_y & 0 \\ k_x s_x & s_y & 0 \\ -k_x p_y s_x - p_x s_x & -p_y s_y - k_y p_x s_y & 1 \end{pmatrix} \qquad (\%\ o7)$$

(% i8) P.K.S.R;

$$\begin{pmatrix} s_x \cos(\theta) - k_y s_y \sin(\theta) & s_x \sin(\theta) + k_y s_y \cos(\theta) & 0 \\ k_x s_x \cos(\theta) - s_y \sin(\theta) & k_x s_x \sin(\theta) + s_y \cos(\theta) & 0 \\ -p_x (s_x \cos(\theta) - k_y s_y \sin(\theta)) - p_y (k_x s_x \cos(\theta) - s_y \sin(\theta)) & -p_y (k_x s_x \sin(\theta) + s_y \cos(\theta)) - p_x (s_x \sin(\theta) + k_y s_y \cos(\theta)) & 1 \end{pmatrix}$$

(% o8)

(% i9) P.K.S.R.T;

$$\begin{pmatrix} s_x \cos(\theta) - k_y s_y \sin(\theta) & s_x \sin(\theta) + k_y s_y \cos(\theta) & 0 \\ k_x s_x \cos(\theta) - s_y \sin(\theta) & k_x s_x \sin(\theta) + s_y \cos(\theta) & 0 \\ -p_x (s_x \cos(\theta) - k_y s_y \sin(\theta)) - p_y (k_x s_x \cos(\theta) - s_y \sin(\theta)) + t_x & -p_y (k_x s_x \sin(\theta) + s_y \cos(\theta)) - p_x (s_x \sin(\theta) + k_y s_y \cos(\theta)) + t_y & 1 \end{pmatrix}$$

(% o9)

This yields the complete Local Transformation. The [final] World Transformation is obtained when the LocalTransformation is multiplied by its container's World Transformation. §